

A Transformer-Based Framework with Dynamic Multiscale Attention Mechanisms for 12-Lead ECG Synthesis

Tanay Chitlur, Nagabhushan Chitlur, Ajay Tripuraneni MD FHRS

COPENHAGEN

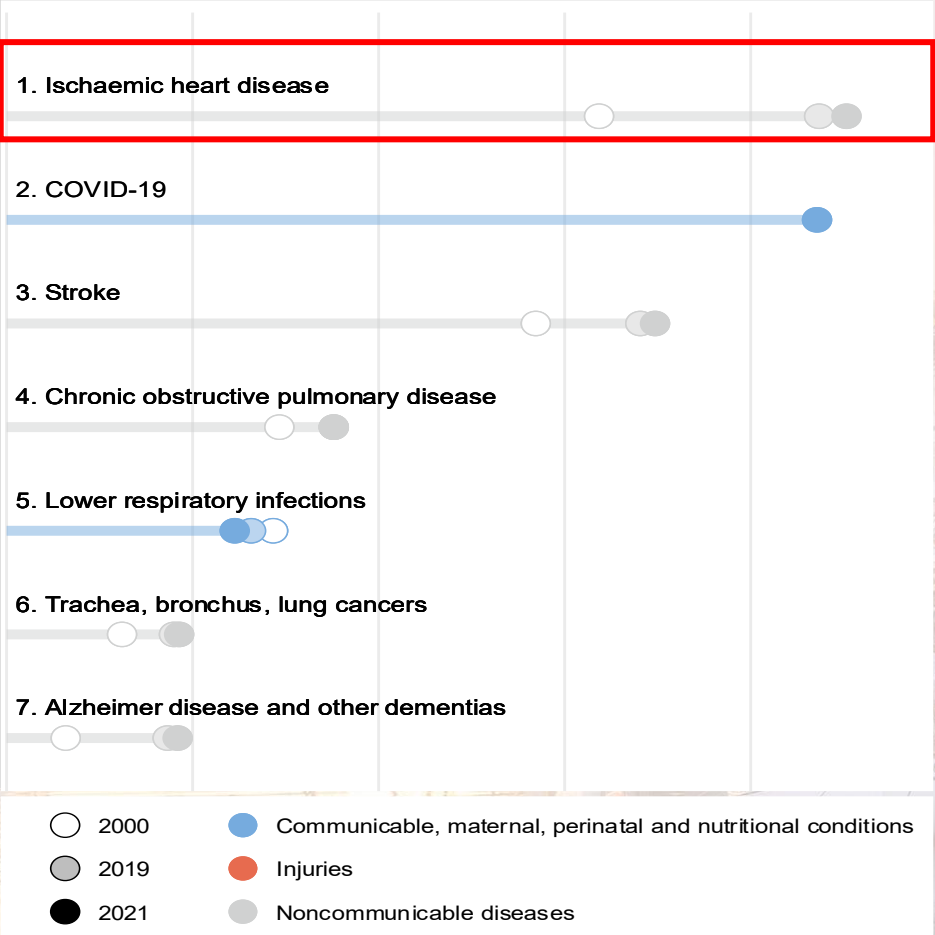


IEEE Engineering
Medicine & Biology Society

47TH ANNUAL
International Conference of the
**IEEE Engineering in Medicine
and Biology Society**

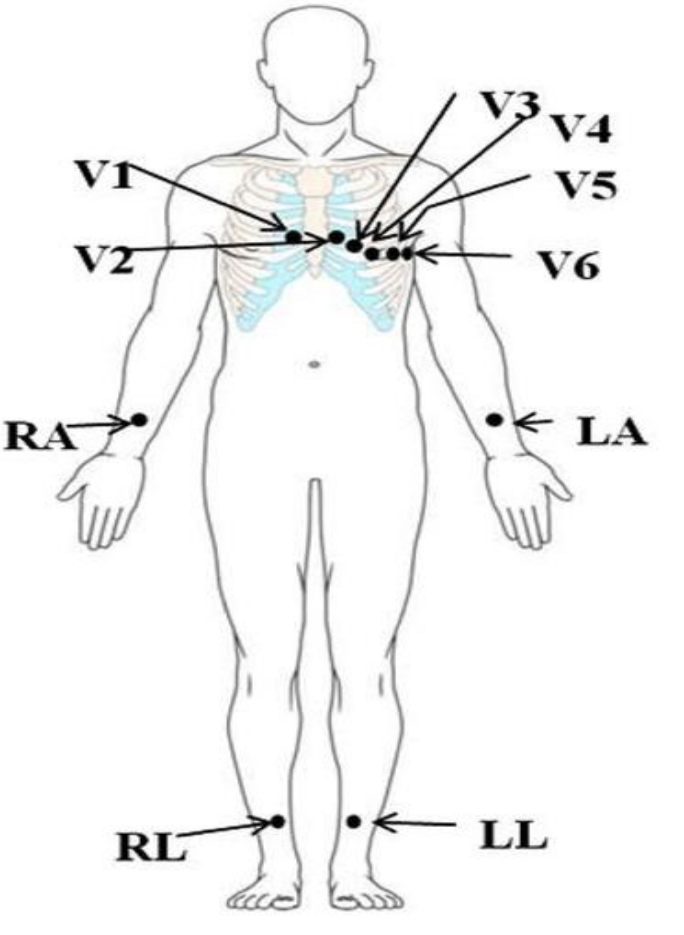
Cardiovascular Disease and Electrocardiograms

Leading Causes of Death Worldwide



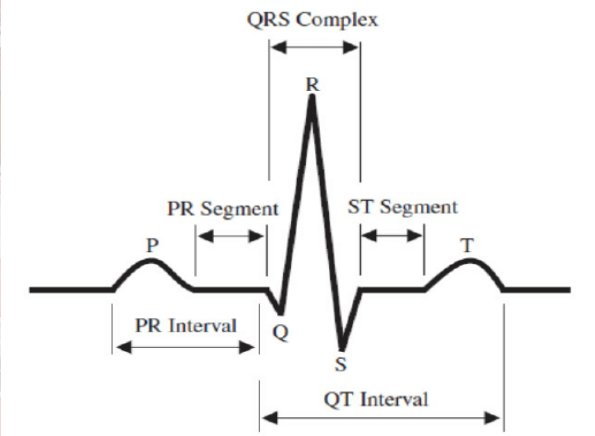
World Health Organization

Leads of an Electrocardiogram (ECG)



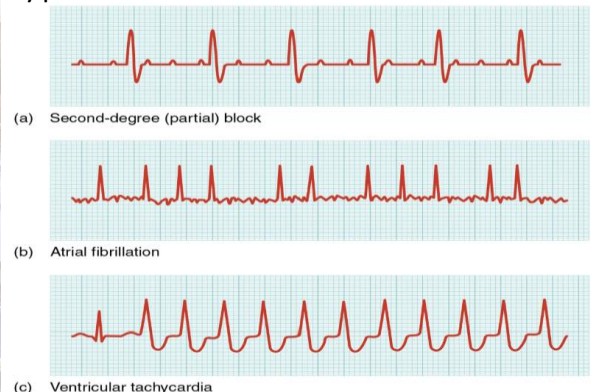
Chaudhari et al., 2021

Parts of a ECG Waveform



Nataraj et al., 2012

Types of ECG Abnormalities



StoryMD, 2025

CVD is the world's leading killer and ECGs are one of the best methods for CVD detection, but most wearables capture only 6 of 12 ECG leads—highlighting the need for new computational methods to synthesize the missing leads

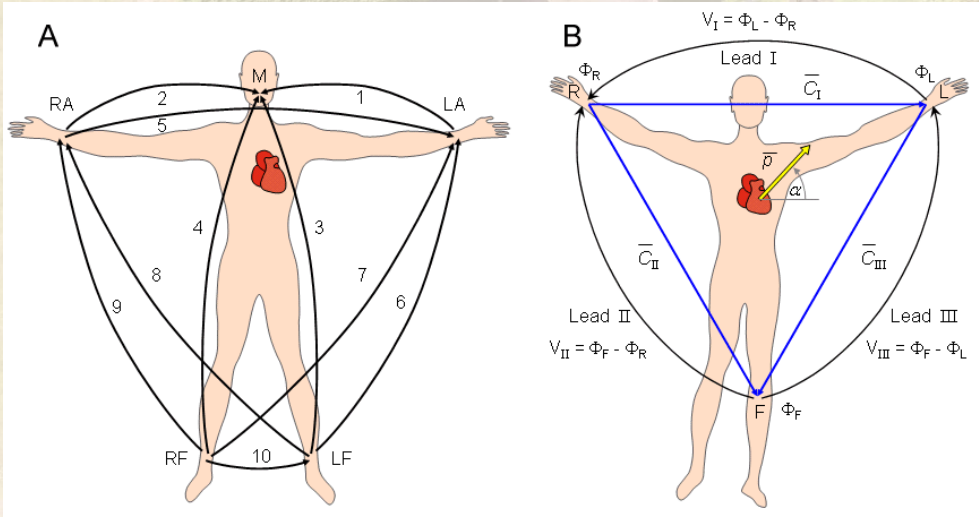
12-Lead Synthesis Strategy

	Lead	Synthesis Method
Input	I	Input Lead
Input	III	Input Lead
Input	V6	Input Lead
Compute	II	Einthoven's Lead System
Compute	aVR	Goldberger's Lead System
Compute	aVL	Goldberger's Lead System
Compute	aVF	Goldberger's Lead System
Synthesize	V1	Synthesize Using Neural Network
Synthesize	V2	Synthesize Using Neural Network
Synthesize	V3	Synthesize Using Neural Network
Synthesize	V4	Synthesize Using Neural Network
Synthesize	V5	Synthesize Using Neural Network

Focus of Paper



van der Zande et al., 2023

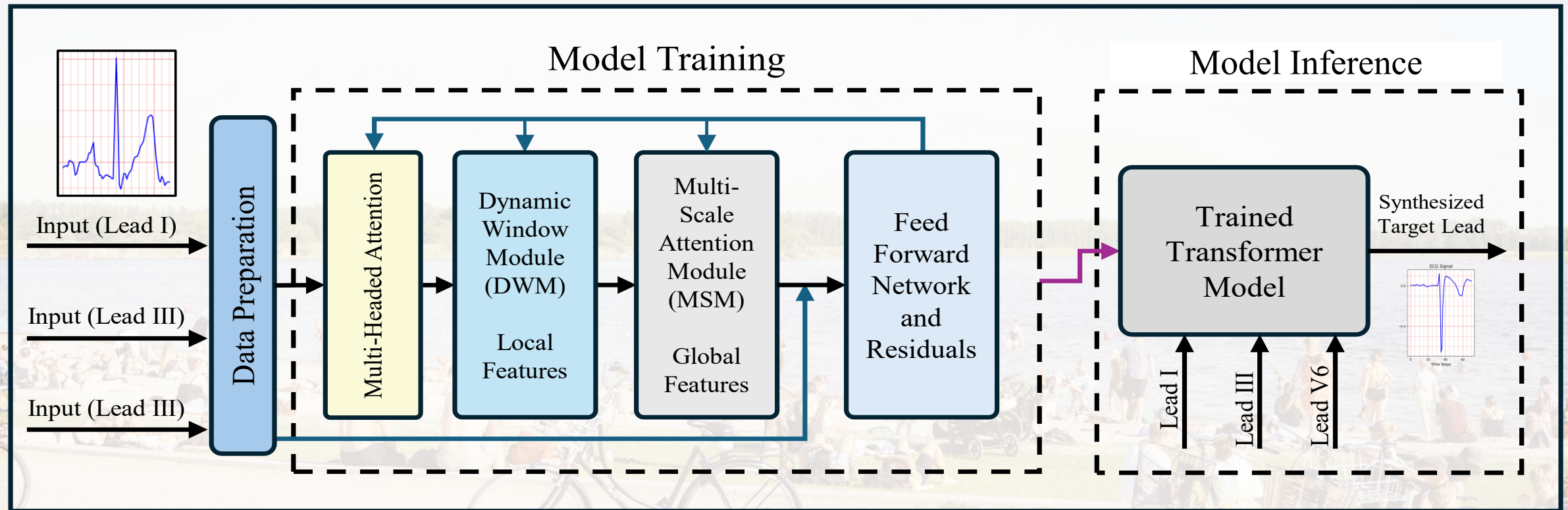


Malmivuo et al., 1975

This work proposes a transformer-based model to synthesize full 12-lead ECGs using just Leads I, III, and V6 as input using existing mathematical models and a new neural network model proposed in this paper

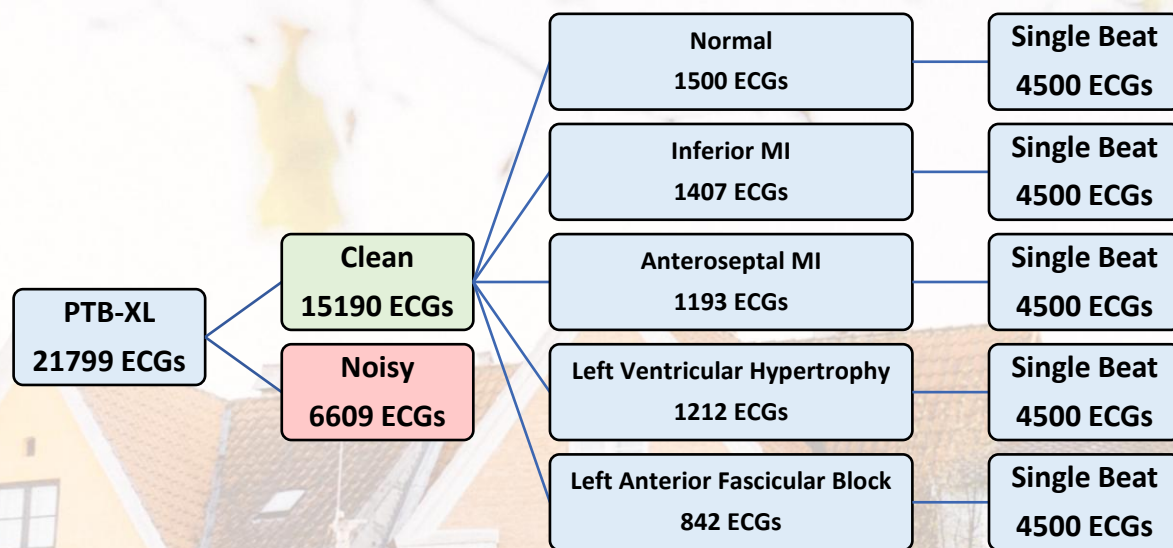
DMTT Model Overview

For each target lead V1, V2, V3, V4, V5



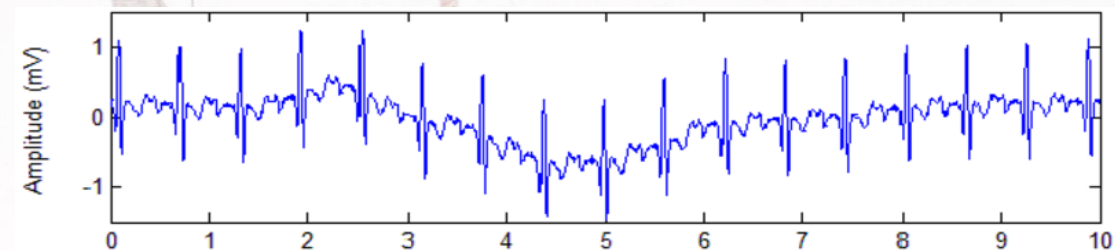
The **Dynamic Multi-Scale Time Series Transformer (DMTT)**, combines dynamic windows and multi-scale context to capture both local and global features—enabling the generation of morphologically accurate ECGs.

Data Processing



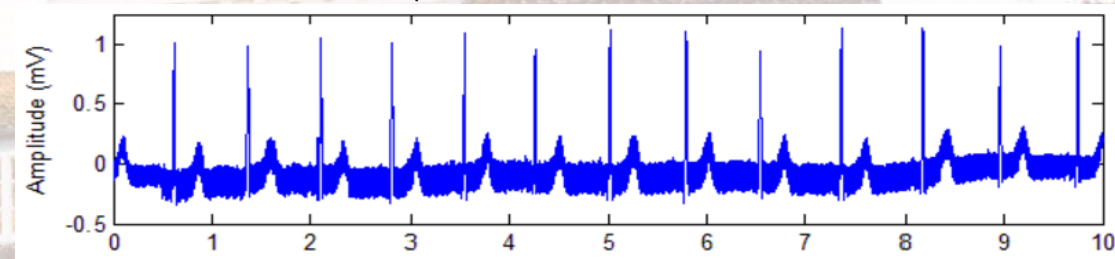
Types of ECG Noise

Baseline wander

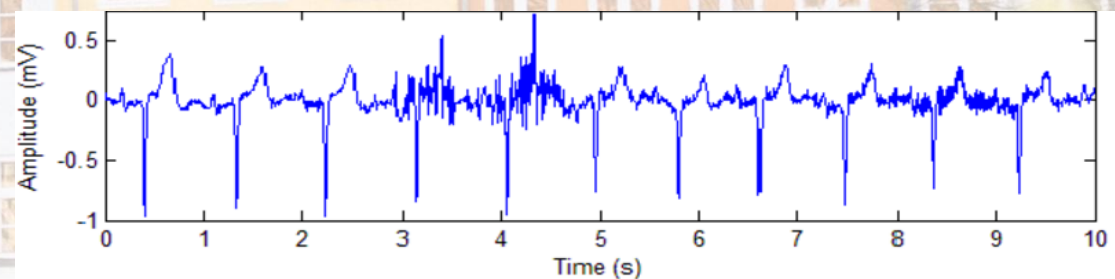


Maggio et al., 2012

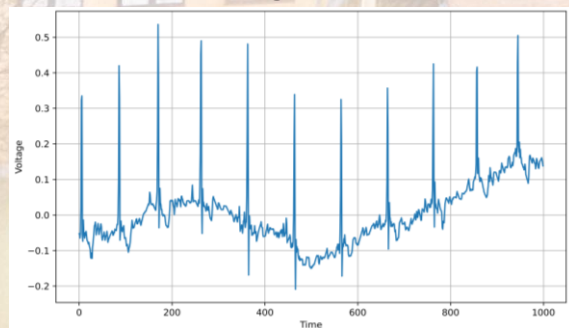
50 Hz power line interference



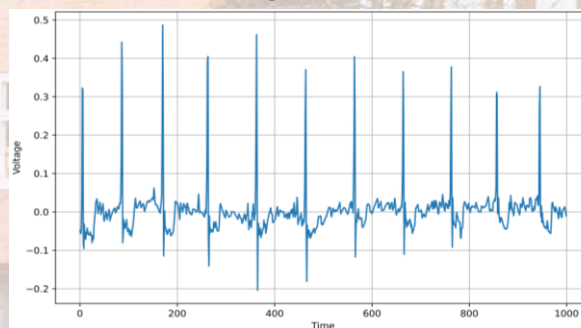
Electromyographic noise



Before High-Pass Filter

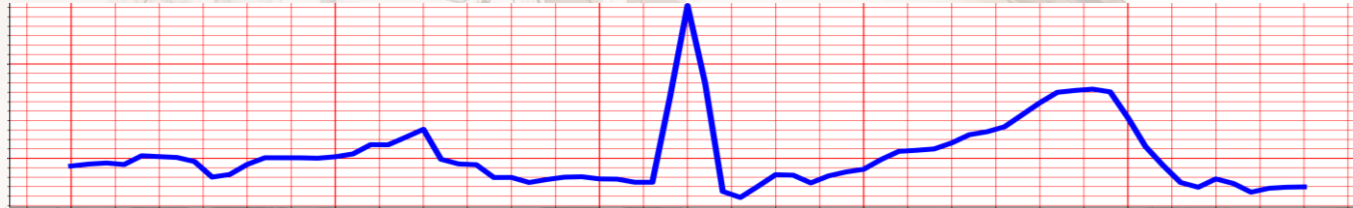


After High-Pass Filter



The final dataset from PTB-XL Dataset consists of 22,500 single beat ECGs from five disease types. Through a process of ECG filtering and selection we arrived at a high-quality ECG samples suitable for model training.

Dynamic Window Module (DWM)



Standard Transformer Attention Windows



Saliency Based Attention Mechanism Windows

- **Feature Extraction & Saliency Mapping:** The model first uses a sub-network to identify high importance features of the P,QRS and T waves generating a “saliency map”
- **Window Resizing Based on Importance:** The DWM adapts the size of the analysis windows based on the importance of the different parts of the signal
- **Emphasizing Key Regions:** The model gives more attention (or weight) to windows that contain highly relevant signal segments, allowing the model to focus on subtle ECG patterns.
- **Output Preparation:** Finally, the selected and emphasized windows are used to create a refined representation of the ECG for better synthesis

Saliency Based Dynamic Window Module (DWM)

Saliency Map Generation

Dense 1: $Z_1 = \text{ReLU}(XW_1 + b_1)$

Dense 2: $Z_2 = \text{ReLU}(Z_1W_2 + b_2)$

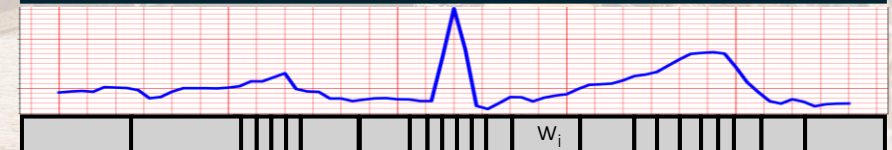
Saliency Map: $s = \sigma(Z_2W_s + b_s)$

$s: 1 \times 71 \times 1$

Refined Feature Map Generation

Window Size: $w = \text{softmax}(sW_w + b_w)$

Refined Feature Map: $F'_i = \text{Pool}_i(F) \odot w_i$



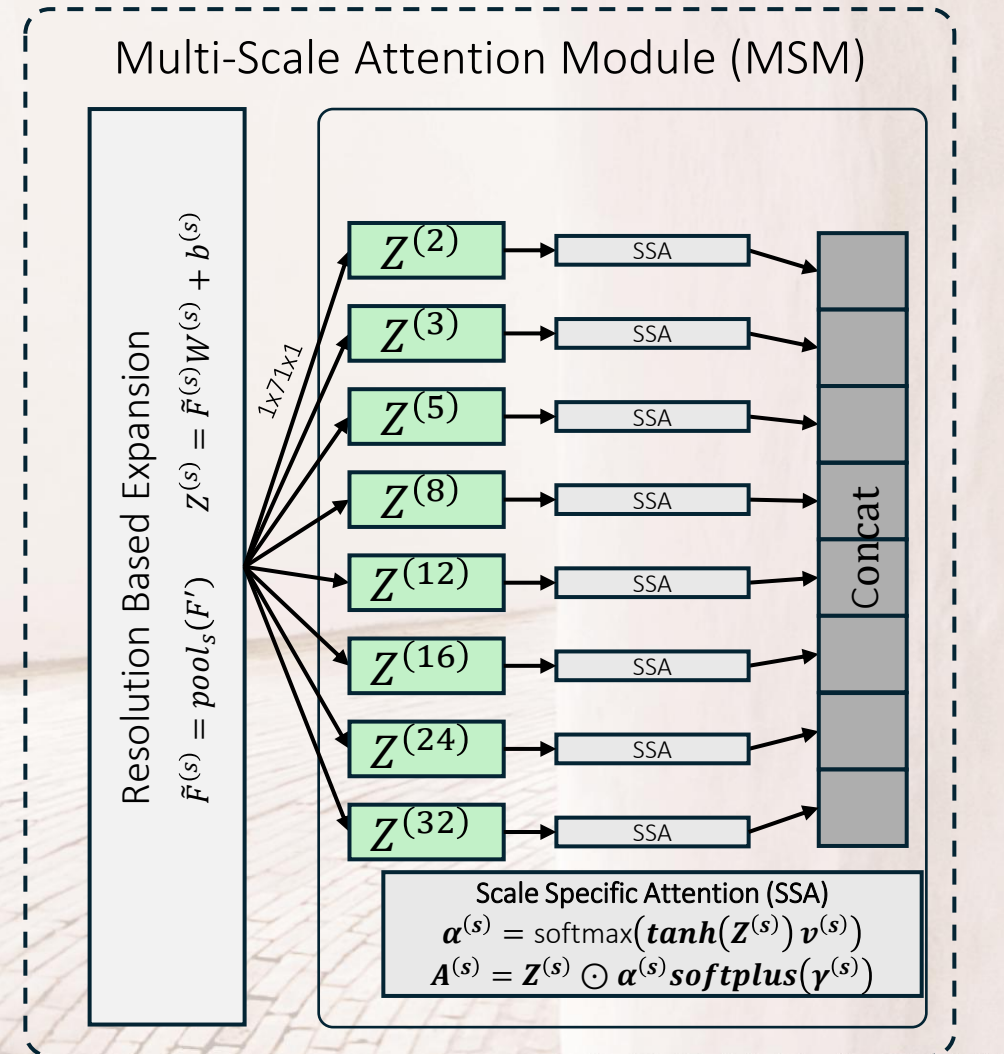
Dynamic Window Sizes Based on Saliency

The Dynamic Window Module adapts window sizes using a learned saliency map, allowing the model to focus more on clinically important ECG segments while filtering out less relevant noise

Multi-Scale Attention Module (MSM)



- **Receives Refined Features:** MSM takes as input the feature map generated by the Dynamic Window Module (DWM), which has local features enhances
- **Multiple Temporal Resolutions:** The MSM looks at the ECG signal using different "window sizes" which happen in parallel
- **Stream Processing in Parallel:** Each window size processes the input in parallel, creating different versions of the signal that highlight patterns at different temporal scales
- **Learned Attention Across Scales:** The module learns which of these scales (or temporal resolutions) are most useful for each part of the ECG
- **Weighted Combination:** It combines the outputs from each scale in a way that emphasizes the most diagnostically valuable ones



The Multi-Scale Module refines output from the DWM by processing the ECG at multiple time resolutions in parallel, re-aligning important wave patterns and improving temporal accuracy

Quantitative Analysis

Leads	AAE of Synthesized Leads V1-V5					
	Normal	ASMI	LAFB	LVH	IMI	Average
Lead V1	0.0425	0.0670	0.0546	0.0643	0.0495	0.0556
Lead V2	0.0769	0.1232	0.1030	0.1228	0.0919	0.1036
Lead V3	0.0759	0.1278	0.1000	0.1232	0.0906	0.1035
Lead V4	0.0524	0.1013	0.0738	0.1010	0.0714	0.0800
Lead V5	0.0317	0.0546	0.0454	0.0572	0.0401	0.0458

$$AAE = \frac{\sum_{i=1}^N |S_{pred,i} - S_{true,i}|}{N}$$

- Absolute difference between the synthesized and real ECG
- Allows direct comparison of synthesized ECGs to original signals, assessing the model's ability to replicate disease-specific and normal patterns
- Measured in Millivolts

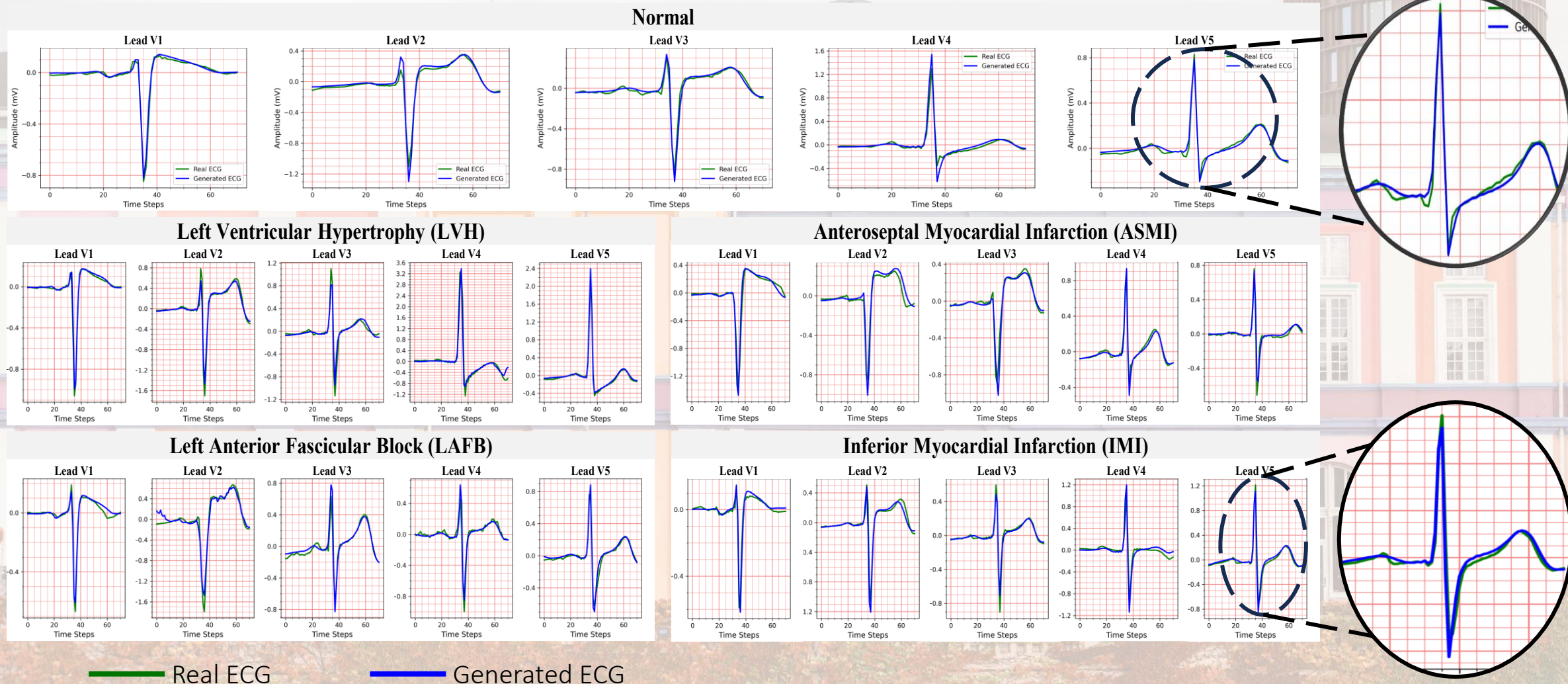
Leads	NRMSE of Synthesized Leads V1-V5					
	Normal	ASMI	LAFB	LVH	IMI	Average
Lead V1	0.0798	0.0956	0.1145	0.0724	0.0940	0.0913
Lead V2	0.0939	0.1402	0.1265	0.1078	0.1082	0.1153
Lead V3	0.0880	0.1483	0.0992	0.1036	0.1007	0.1079
Lead V4	0.0541	0.1462	0.0729	0.0777	0.0802	0.0862
Lead V5	0.0345	0.0852	0.0516	0.0443	0.0476	0.05264

$$NRMSE = \frac{\sqrt{\sum_{i=1}^N \frac{(S_{pred,i} - S_{true,i})^2}{N}}}{S_{pred,max} - S_{pred,min}}$$

- Difference between the synthesized ECG and actual ECG while accounting for the data scale through normalization
- For comparing the accuracies across different leads, conditions, and datasets

DMTT shows strong performance across both AAE and NRMSE, effectively capturing ECG morphology even in complex regions — highlighting its viability

Qualitative Analysis



Synthesized ECGs closely match real signals, accurately capturing P, QRS, and T wave morphology across all disease types

Conclusions and Future Work

- Demonstrated that realistic ECG signals can be synthesized using only three input leads (I, III, and V6)
- Unique combination of dynamically adapting windows and multi-scale attention captures local and global features
- Quantitative and qualitative analysis of synthesized ECGs shows model is accurate for beat morphology and timing
- 12-lead ECGs can be generated using only leads recorded from wearable devices, aiding at-home-monitoring
- Future Work - Incorporate individual biometrics (e.g. age, gender, race, weight) to personalize ECG synthesis